Question 16 **1 mark**

Identify the expression that is equivalent to $\tan\theta\left(\csc^2\theta - 1\right)$.

a. $\cot\theta$

b. $\tan\theta$

c. $\sec\theta$

d. $\sin\theta$

Question 17 **1 mark**

Identify the expression that represents the number of ways five books can be arranged on a shelf if two of them must be together.

a. $5!2!$

b. $5!-2!$

c. $4!2!$

d. $3!2!$

Question 18 **1 mark**

Identify the value of x in the equation, $\log_8 x = -\frac{1}{3}$.

a. -2

b. $-\frac{1}{2}$

c. $\frac{1}{2}$

d. 2

Question 19 — 1 mark

Given the functions, $f(x)=\dfrac{1}{x}$ and $g(x)=\sqrt{x-1}$, identify the domain of $f(g(x))$.

a. $(-\infty,\infty)$

b. $(-\infty,1)\cup(1,\infty)$

c. $[1,\infty)$

d. $(1,\infty)$

Question 20 — 1 mark

Given $f(x)=\dfrac{3(x+2)}{5(x+2)(x-2)}$, identify the equation of the horizontal asymptote.

a. $y=-2$

b. $y=0$

c. $y=\dfrac{3}{5}$

d. $y=2$

Question 21 — 1 mark

Identify the range of the sinusoidal function, $f(x)=2\sin x+3$.

a. $\{y \mid y\in\mathbb{R}\}$

b. $\{y \mid 2\le y\le 4,\ y\in\mathbb{R}\}$

c. $\{y \mid -2\le y\le 2,\ y\in\mathbb{R}\}$

d. $\{y \mid 1\le y\le 5,\ y\in\mathbb{R}\}$

Question 22 **1 mark**

Given that $(x-3)$ is a factor of the polynomial $P(x)$, identify which of the following is true.

a. $P(3)=0$

b. $P(0)=3$

c. $P(0)=-3$

d. $P(-3)=0$

Question 23 **1 mark**

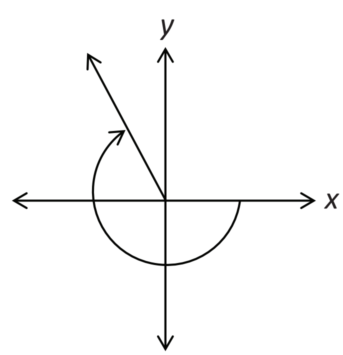
Identify the value that best represents the angle in standard position.

a. −4.4 radians

b. −2 radians

c. 1.3 radians

d. 1.9 radians



Question 24 **1 mark**

Identify the expression that is equivalent to $\log_a(x+2)+3\log_a(x)$.

a. $\log_a\left(3(x+2)\right)$

b. $\log_a\left(x^3(x+2)\right)$

c. $\log_a\left(\frac{(x+2)}{x^3}\right)$

d. $\log_a x+\log_a 2+\log_a x^3$

Question 25 **3 marks** 117

Evaluate.

$$\tan^3\left(\frac{3\pi}{4}\right)+\csc\left(\frac{-4\pi}{3}\right)\cos\left(\frac{13\pi}{6}\right)$$

Question 26 **2 marks** 118

If $a = \log 4$ and $b = \log 3$, express $\log 36$ in terms of a and b.

Question 27 **2 marks** 119

Justify that $\tan 2x$ has a non-permissible value at $x = \frac{\pi}{4}$.

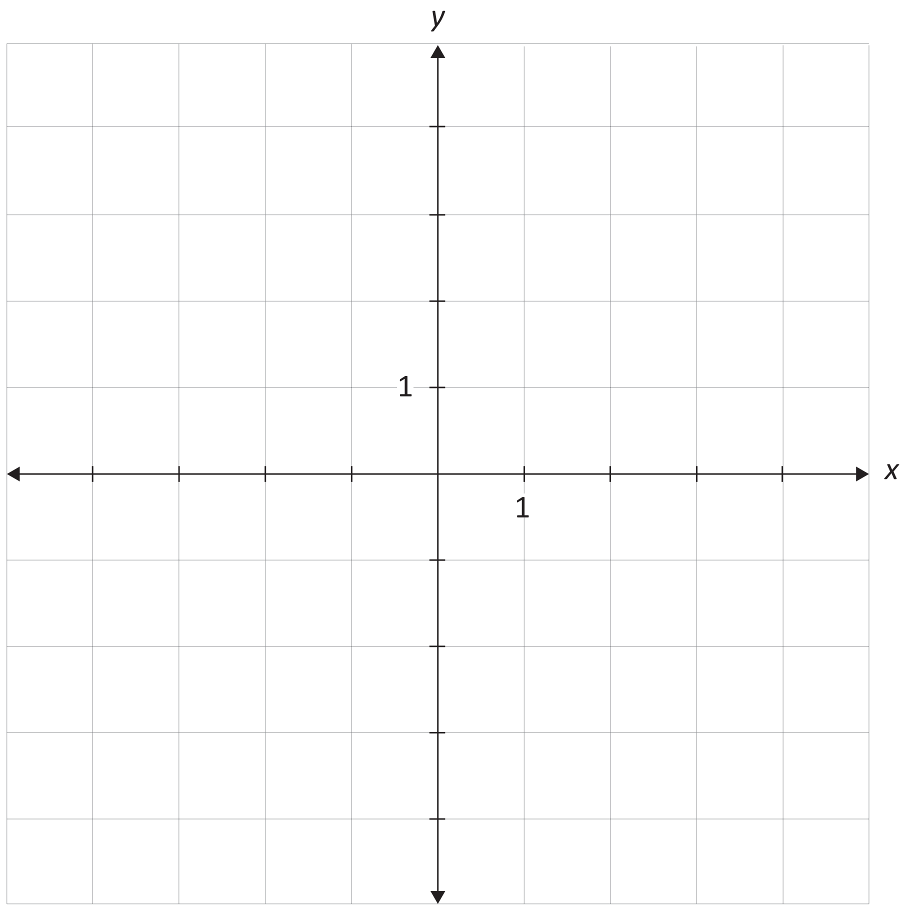
Question 28 — 2 marks 120

Determine the exact value of $\tan\theta$, if $\sec\theta = \frac{6}{5}$ and θ terminates in quadrant IV.

Question 29 — a) 2 marks b) 1 mark 121 122

Given the function, $g(x) = \dfrac{4}{x^2+1}$,

a) sketch the graph of $g(x)$.



b) state the range of the graph of $g(x)$.

Question 30 — a) 2 marks b) 1 mark 123 124

Given $y = x^2 - 4$,

a) determine the equation of the inverse.

$y =$ ______________________________________

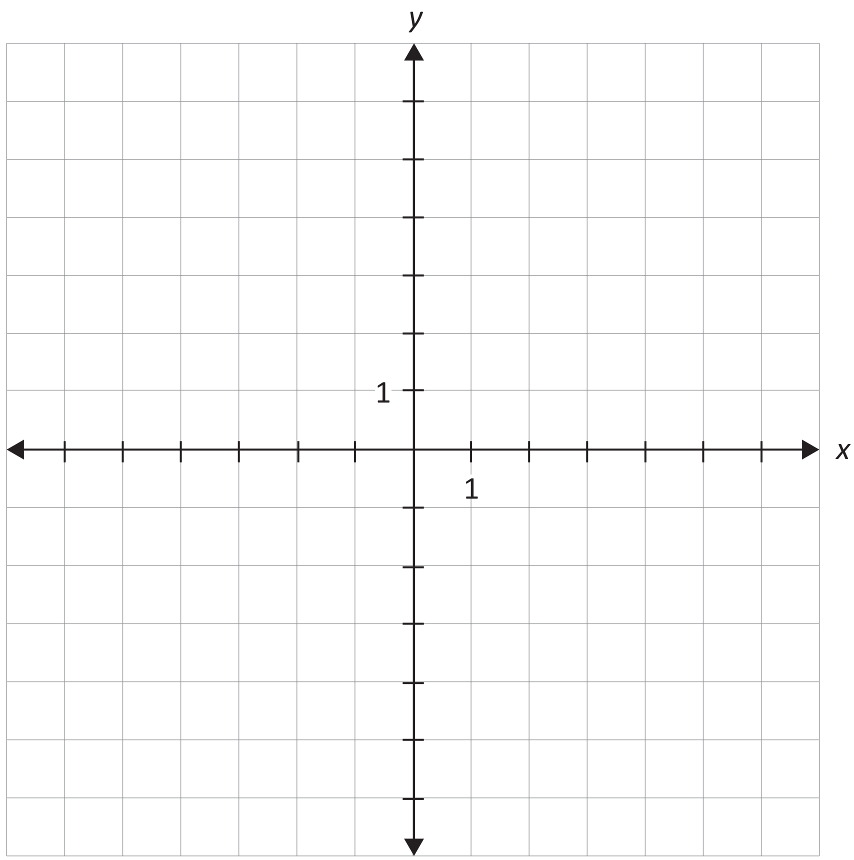
b) state a restriction on the domain of $y = x^2 - 4$, in order for its inverse to be a function.

Question 31 3 marks 125

Given $\cos\alpha = -\frac{1}{3}$ and $\sin\beta = -\frac{2}{3}$ where α and β terminate in the same quadrant, determine the exact value of $\sin(\alpha - \beta)$.

Question 32 **4 marks** 126

Sketch the graph of $y+2=-\sqrt{4x}$.



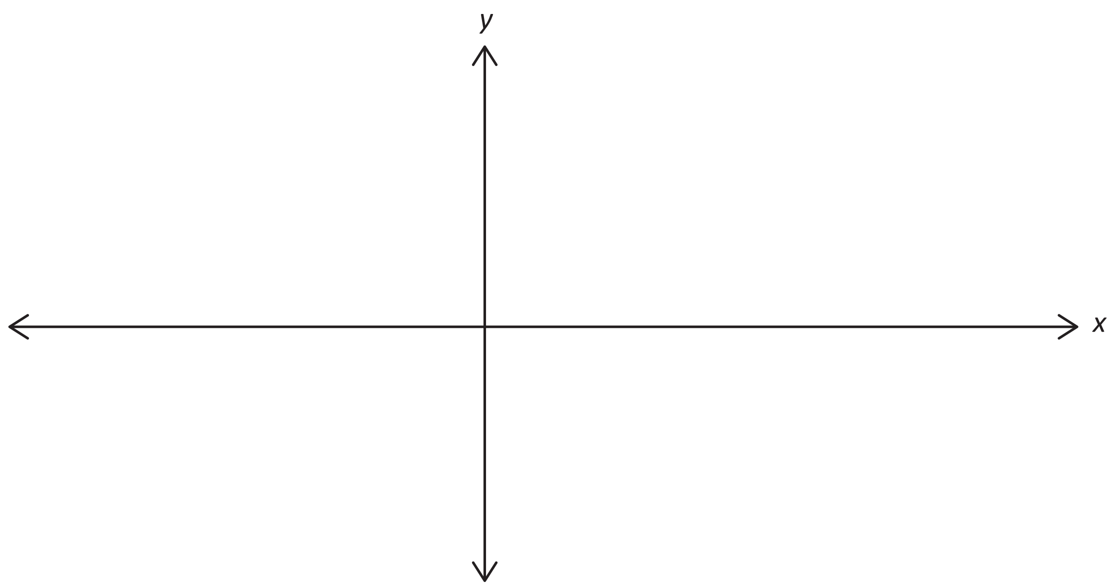
Question 33 — 3 marks 127

Express $p(x) = x^4 - x^3 - 6x^2 + 4x + 8$ in completely factored form, given that $(x+1)$ is a factor of $p(x)$.

$p(x) =$ ____________________________________

Question 34 **4 marks** 128

Sketch at least one period of the graph of $y = -\sin\left(2\left(x - \frac{\pi}{4}\right)\right)$.

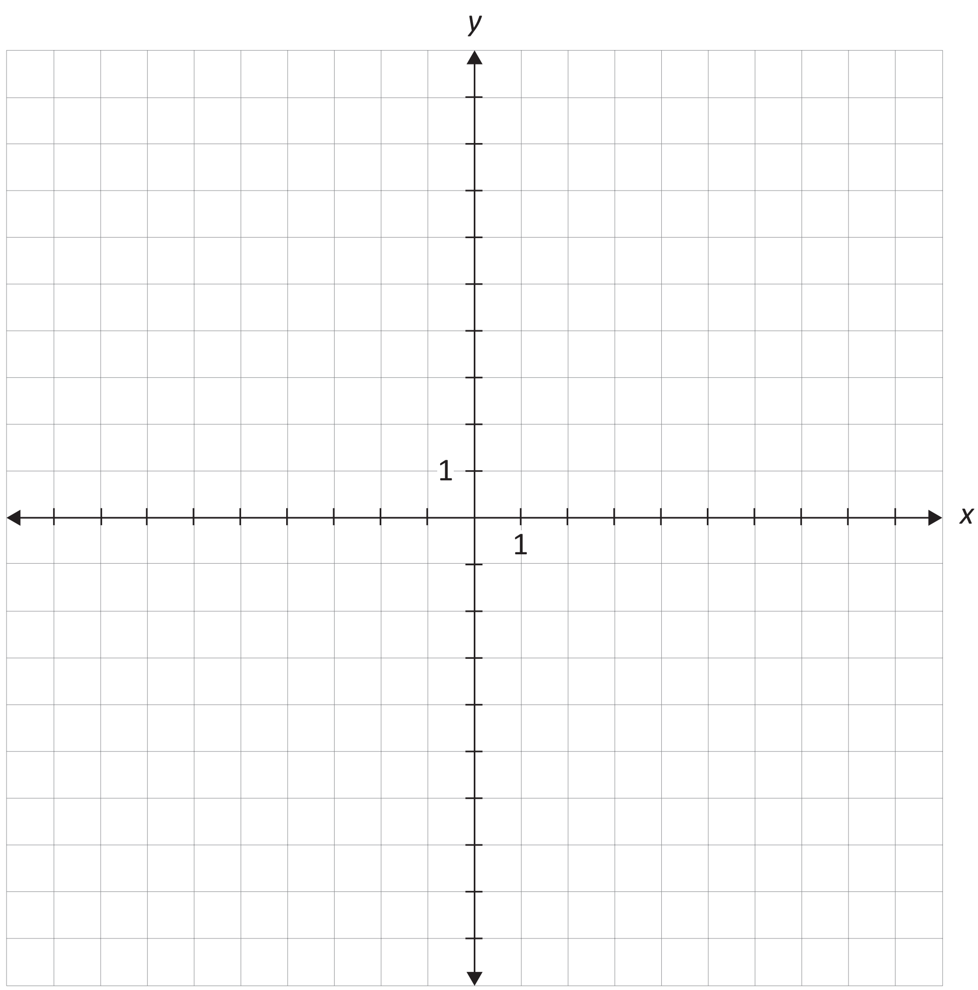


Question 35 — a) 1 mark b) 2 marks 129 130

Given $g(x) = \log_2(x) - 3$,

a) determine the x-intercept of the graph.

b) sketch the graph of $g(x)$.



Question 36 **1 mark** 131

Modesola incorrectly evaluated the expression, $\cos^2\left(30^\circ\right)-\sin^2\left(30^\circ\right)$.

Modesola's answer:

$$\cos^2(30^\circ) - \sin^2(30^\circ)$$

$$= 2\cos(30^\circ)$$

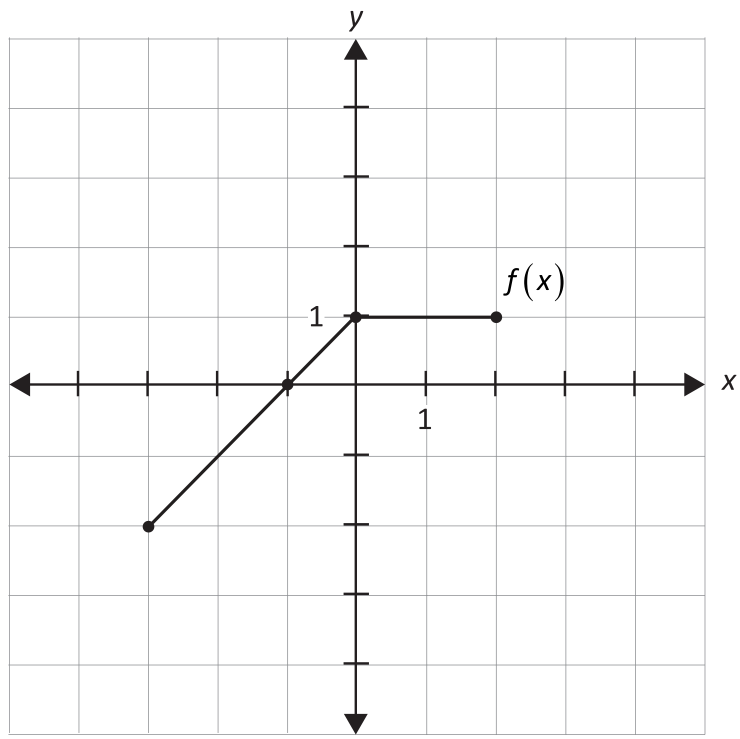
$$= 2\left(\frac{\sqrt{3}}{2}\right)$$

$$= \sqrt{3}$$

Describe their error.

Question 37 **1 mark** 132

Given the graph of $y = f(x)$, state the range of $y = \sqrt{f(x)}$.



Question 38 — 1 mark 133

Determine a value of *x* that satisfies $\sin^2(2x-1)+\cos^2(x+5)=1$.

Question 39 **2 marks** 134

Solve, algebraically.

$$9^{2x+3} = \left(\frac{1}{3}\right)^{x+1}$$

Question 40 **2 marks** 135

Determine the total number of arrangements of the letters in the word GUIDES if the vowels and consonants must alternate.

Question 41 **2 marks** 136

Solve, algebraically.

$$\frac{5!(n-5)!}{3!(n-6)!}=80$$

Question 42 **1 mark** 137

Justify that 2.1 is a better estimate than 2.6 for the value of $\log_5 29$.

Question 43 2 marks 138

When $p(x) = x^3 + kx + 6$ is divided by $(x+2)$, the remainder is 4. Determine the value of k.

Question 44 **3 marks** 139

Describe the transformations used to obtain the graph of the function $y = -f\left(7x + 14\right)$ from the graph of $y = f\left(x\right)$.